Associations of whole-blood fatty acids and dietary intakes with prostate cancer in Jamaica.

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OBJECTIVE: To investigate the association of whole-blood fatty acids and reported intakes of fats with risk of prostate cancer (PCa). DESIGN: Case-control study of 209 men 40-80 years old with newly diagnosed, histologically confirmed prostate cancer and 226 cancer-free men attending the same urology clinics. Whole-blood fatty acid composition (mol%) was measured by gas chromatography and diet assessed by food frequency questionnaire. RESULTS: High whole-blood oleic acid composition (tertile 3 vs. tertile 1: OR, 0.37; CI, 0.14-0.0.98) and moderate palmitic acid proportions (tertile 2: OR, 0.29; CI, 0.12-0.70) (tertile 3: OR, 0.53; CI, 0.19-1.54) were inversely related to risk of PCa, whereas men with high linolenic acid proportions were at increased likelihood of PCa (tertile 3 vs. tertile 1: OR, 2.06; 1.29-3.27). Blood myristic, stearic and palmitoleic acids were not associated with PCa. Higher intakes of dietary MUFA were inversely related to prostate cancer (tertile 3 vs. tertile 1: OR, 0.39; CI 0.16-0.92). The principal source of dietary MUFA was avocado intake. Dietary intakes of other fats were not associated with PCa. CONCLUSIONS: Whole-blood and dietary MUFA reduced the risk of prostate cancer. The association may be related to avocado intakes. High blood linolenic acid was directly related to prostate cancer. These associations warrant further investigation.